

Unit-5

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Research Ethics

- Research ethics refers to the **moral principles, professional standards, and responsible practices** that guide researchers during:
 - Planning research
 - Conducting research
 - Collecting data
 - Analyzing data
 - Reporting findings
 - Publishing research

Why Research Ethics is Important

Ethical research ensures:

- Participants are treated with dignity and respect.
- Participation is voluntary.
- Risks to participants are minimized.
- Privacy and confidentiality are protected.
- Research data are handled responsibly.
- Findings are reported honestly.
- Sources are properly acknowledged.
- Plagiarism and research misconduct are avoided.

Working with Human Subjects

Human-Subject Research

- Research involving people as participants is called **human-subject research**.

Examples: Interviews , Questionnaires , Surveys , Usability testing , HCI experiments , Focus groups , User observation , Behavioral studies , Online-user studies

In HCI research, people are commonly involved in usability studies, focus groups, experiments, and ethnographic research.

Responsibilities When Working with Participants

Researchers should:

- Respect participants' rights.
- Explain the purpose of the study.
- Obtain appropriate consent.
- Minimize risks.
- Protect personal information.
- Maintain confidentiality.
- Allow voluntary participation.
- Follow applicable ethical-review requirements.
- Report participant data honestly.

Fundamental Principles of Research Ethics

Three important principles are:

Respect for Persons

- Recognize participants as autonomous individuals.

Beneficence

- Maximize benefits and minimize risks.

Justice

- Ensure fair selection and treatment of participants.

These principles provide a framework for ethical research involving human participants.

Respect for Persons

Respect for persons means allowing individuals to make informed and voluntary decisions about participation.

Researchers should:

- Provide adequate information.
- Allow voluntary participation.
- Avoid unnecessary pressure.
- Respect refusal to participate.
- Respect withdrawal procedures.
- Provide additional protection to vulnerable participants.

Example: A student should understand the purpose and requirements of a usability study before agreeing to participate.

Beneficence

Beneficence means: Maximize possible benefits while minimizing possible harm.

Possible Risks

- Physical discomfort
- Psychological stress
- Embarrassment
- Loss of privacy
- Social consequences
- Disclosure of personal information

Justice

Justice means that the **benefits and burdens of research should be distributed fairly.**

Researchers should avoid selecting participants simply because they are:

- Easily accessible
- Economically disadvantaged
- Powerless
- Dependent on the researcher
- Unable to refuse participation

Example: A system designed for older adults should be evaluated with appropriate older-adult participants rather than only convenient university students.

PARTICIPANT PROTECTION

Informed Consent

- **Informed consent** is the process through which participants receive sufficient information about a research study and voluntarily agree to participate.
- **Participants should know:**
 - Purpose of research
 - Activities involved
 - Expected duration
 - Possible risks
 - Possible benefits
 - Data collection procedures
 - Data usage
 - Privacy arrangements
 - Withdrawal procedures
 - Contact information

Example of Consent

"You are invited to participate in a study evaluating the usability of a university learning-management system."

Participants should be informed that:

- Participation is voluntary.
- They can ask questions.
- Their information will be handled appropriately.
- Research data will be used for the stated purpose.
- Applicable withdrawal procedures will be explained.

Key Idea : Informed + Voluntary + Understandable = Meaningful Consent

Privacy and Confidentiality

Privacy

- Privacy concerns an individual's **control over personal information and activities.**

Confidentiality

- Confidentiality means **protecting information obtained from participants from unauthorized disclosure.**

Actual Participant	Research Record
Ram Sharma	P001
Sita Thapa	P002
Hari KC	P003

Privacy	Confidentiality
Concerned with personal information	Concerned with protection of collected information
Participant controls access	Researcher protects information
Focuses on what may be collected	Focuses on how collected data are handled
Example: limiting unnecessary questions	Example: securing participant records

Protection from Harm

- Researchers should consider different types of harm.

Physical

- Injury
- Fatigue
- Discomfort

Psychological

- Anxiety
- Stress
- Embarrassment

Social

- Discrimination
- Reputation damage

Privacy

- Unauthorized disclosure
- Data exposure

Vulnerable Participants

Some participants may require additional protection.

Examples

- Children
- People with limited decision-making capacity
- Patients
- People with disabilities
- Economically disadvantaged populations

- People in dependent relationships

Researchers should:

- Use understandable information.
- Provide appropriate consent procedures.
- Ensure accessibility.
- Avoid unnecessary risks.
- Provide additional safeguards where required.

Deception in Research

Deception occurs when researchers intentionally withhold or provide incomplete information about some aspect of a study.

Possible Use: Sometimes complete disclosure may influence participant behavior.

Ethical Requirements

Deception should:

- Have strong research justification.
- Minimize risk.
- Not unnecessarily distress participants.
- Meet applicable ethical requirements.
- Be followed by appropriate explanation/debriefing where applicable.

Ethical Review

- Research involving human participants may require review by an:
- **Institutional Review Board (IRB)**
- **Research Ethics Committee (REC)**

Review May Consider

- Participant risks
- Consent procedures
- Privacy
- Confidentiality
- Recruitment
- Data management
- Research procedures

RESEARCH MISCONDUCT

Major forms of research misconduct include:

Fabrication

- Creating data or results that were never obtained.

Falsification

- Manipulating research materials, procedures, or results.

Plagiarism

- Presenting another person's work, ideas, words, or data as one's own without proper acknowledgment.

Plagiarism

Using another person's:

- Words
- Ideas
- Data
- Figures
- Tables
- Research findings
- without appropriate acknowledgment.

Types

- Direct plagiarism
- Mosaic plagiarism
- Improper paraphrasing
- Self-plagiarism
- Uncredited figures/tables
- Uncredited ideas

How to Avoid Plagiarism

Researchers should:

- Take accurate notes.
- Record source information.
- Cite sources while writing.
- Use quotation marks for direct quotations.
- Paraphrase genuinely.
- Acknowledge borrowed ideas.
- Reference figures and tables.
- Check the final document for plagiarism.

Golden Rule : If an idea or material comes from another source, acknowledge the source.

Authorship Ethics

Authorship should reflect meaningful contribution.

Researchers should avoid:

Gift Authorship

- Adding someone who did not make an appropriate contribution.

Ghost Authorship

- Excluding someone who made a substantial contribution.

Other Issues

- Unfair author ordering
- Claiming another person's work
- Misrepresenting contributions

Principle : Authorship should be fair, transparent, and contribution-based.

Conflict of Interest

A **conflict of interest** occurs when personal, financial, professional, or institutional interests could influence—or appear to influence—the research.

Example

- A researcher evaluates a software product developed by a company in which the researcher has a financial interest.

Good Practice

- Identify conflicts.
- Disclose relevant conflicts.
- Follow institutional/publication policies.

Research Proposal?

A **research proposal** is a formal document that explains:

- What the researcher wants to study.
- Why the study is important.
- What previous research says.
- What gap exists.
- What objectives will be addressed.
- How the research will be conducted.
- What outcomes are expected.

Uses

A proposal is commonly used for:

- Research planning
- Academic approval
- Registration
- Ethical approval
- Funding applications

Purpose of a Research Proposal

A research proposal helps researchers to:

- Define the research problem.
- Establish research significance.
- Review previous research.
- Identify research gaps.
- Define objectives.
- Formulate research questions/hypotheses.
- Plan methodology.
- Identify expected outcomes.
- Estimate time and resources.
- Obtain approval or funding.

Components of a Research Proposal

- Title
- Introduction
- Background
- Problem Statement
- Research Gap
- Motivation
- Research Questions
- Objectives
- Hypothesis
- Literature Review
- Methodology
- Research Design
- Population and Sample
- Data Collection
- Data Analysis
- Ethical Considerations
- Expected Outcomes
- Work Plan
- Budget/Resources
- References
- Appendices

Research Problem and Research Gap

Research Problem

- A research problem is a **clearly defined issue, difficulty, gap, or unanswered question** requiring systematic investigation.

Research Gap

- A research gap is an **unresolved issue or inadequately studied area** in existing research.

A Gap May Exist Because:

- Previous results conflict.
- A population has not been studied.
- Existing methods have limitations.
- Technology has changed.
- Existing datasets are inadequate.
- A specific context has not been investigated.

Research Objectives

- Objectives describe **what the research intends to achieve.**

Good Objectives Should Be:

- Specific
- Measurable
- Achievable
- Relevant
- Clearly related to the research problem

Example Topic: AI-Based
Devanagari Numeric Symbol
Recognition

Objectives:

- Develop a dataset of Devanagari numeric symbols.
- Preprocess collected images.
- Develop a CNN-based recognition model.
- Evaluate model performance.
- Compare the proposed model with existing approaches.

Literature Review in a Proposal

A literature review:

- Summarizes previous research.
- Critically evaluates existing studies.
- Identifies methods used.
- Compares findings.
- Identifies limitations.
- Establishes the research gap.

Literature Review Should Answer

- **What is already known, what is not known, and what will this research contribute?**

Literature Review Comparison

Author	Method	Major Finding	Limitation	Gap
A	CNN	High accuracy	Small dataset	Larger dataset
B	SVM	Good performance	Feature-dependent	Deep learning
C	CNN	Better accuracy	Limited classes	More diverse data

Key Point : A literature review should **analyze and compare**, not simply list previous studies.

Methodology

- Methodology explains **how the research will be conducted.**

It May Include:

- Research approach
- Research design
- Population
- Sample
- Variables
- Data sources

- Data collection
- Research instruments
- Data analysis
- Ethical procedures

Common Approaches

- **Qualitative, Quantitative, Mixed Methods**

Data Collection

Common Data Collection Methods

- Observation
- Personal interview
- Telephone interview
- Questionnaire
- Schedule
- Experiment
- Existing datasets/documents

Selection Depends On:

- Research problem
- Research objectives
- Type of data required
- Available time
- Cost and resources

Expected Outcomes

Expected outcomes describe what the researcher expects to produce.

Examples

- New algorithm
- Improved accuracy
- Validated model
- Framework
- Dataset

- Design guidelines
- Practical recommendations
- Comparative findings
- Theoretical contribution

Principle

Expected outcomes must be realistic and connected to the research objectives.

Research Work Plan

Activity/Duration	M1	M2	M3	M4	M5
Topic Selection	✓				
Literature Review	✓	✓			
Methodology		✓	✓		
Data Collection			✓	✓	
Analysis				✓	
Report Writing				✓	✓

What is a Research Report?

A **research report** is a systematic • Discussion
written presentation of: • Conclusions

- Research problem
- Objectives
- Methodology
- Data
- Analysis
- Findings
- Recommendations

Purpose

- To communicate research findings and provide a permanent record of the research process.

Purpose of a Research Report

A research report helps to:

- Communicate research findings.
- Document the research process.
- Provide evidence for conclusions.
- Allow others to evaluate the study.
- Add knowledge to the field.
- Support future research.
- Support practical decision-making.
- Provide an academic record.

Types of Research Reports

Common forms include:

- Research paper
- Journal article
- Conference paper
- Technical report
- Thesis
- Dissertation
- Project report
- Book
- White paper
- Oral presentation

Major Components of a Research Report

Preliminary Pages

- Cover Page
- Title Page
- Declaration/Certificate
- Acknowledgement
- Abstract
- Table of Contents
- List of Tables
- List of Figures
- Abbreviations

Main Text

- Introduction
- Literature Review
- Methodology
- Results
- Discussion
- Conclusion

End Matter

- References
- Bibliography
- Appendices
- Glossary
- Index

Introduction Chapter

The introduction establishes the context of the research.

Main Contents

- Background
- Research problem
- Research questions
- Objectives
- Hypothesis
- Scope
- Significance
- Limitations

Main Question : Why is this research necessary?

Methodology Chapter

The methodology explains **what was actually done**.

Contents

- Research design
- Population
- Sample
- Variables
- Data sources
- Instruments
- Data collection procedure
- Data processing
- Data analysis
- Ethical considerations

Main Question

How was the research conducted?

Results and Analysis

Results

- Present what was obtained from the analysis.

Common Presentation Methods

- Tables
- Charts
- Graphs
- Diagrams
- Statistical summaries

Important

- **Results = What the analysis produced**
- **Discussion = What the results mean**

Discussion

The discussion interprets the findings.

It answers:

- What do the results mean?
- Were the objectives achieved?
- Was the hypothesis supported?
- How do results compare with previous studies?
- Why might the results differ?
- What are the implications?

Key Point : The discussion should be based on **evidence**, not personal opinion.

Conclusion and Recommendations

Conclusion

Summarizes:

- Major findings
- Answers to research questions
- Achievement of objectives
- Main contribution

Recommendations

Suggest:

- Practical improvements
- Policy implications
- Future research
- Improvements to existing methods

Important

Conclusions should be supported by the actual research findings.